A Conditional Coefficient-Only Obstruction for a Fixed Theta Response Majorant

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Abstract

A single finite certification attempt at shift 1/128 for a minimum theta-weighted operator stopped at a nonpositive factor pivot. We separate that failed production from two successful subsequent exact diagnostics. The first supplies one dyadic direction in which the saved unpadded center already fails the necessary response inequality. The second bounds the largest improvement possible by changing the coefficient matrix alone. For the fixed observation and trial maps, action approximants, norm and error bounds, trial-Gram majorant, shift and Young parameter, the bound is strictly smaller than the recorded excess. Thus every coefficient matrix in this fixed family fails that direction, both for the nominal polynomial and, conditionally, for the physical upper-majorant family. The nominal conclusion is also conditional: its positive normal-matrix bound uses an enclosure bridge from inherited analytic and physical premises. The argument is elementary square completion with exact saved-data verification. It gives neither a negative direction of the true operator nor a general impossibility at shift 1/128. Source inclusions are not regenerated, and no conclusion about the Riemann hypothesis is obtained.

1 Question and scope

The earlier fixed-threshold calculation and its subsequent small fixed-coefficient improvement concern the same minimum theta realization [1, 2]. The present result concerns a different, unsuccessful attempt at $\delta = 1/128$. Failure of one coefficient proposal does not exclude other coefficients. Failure of an upper bound does not determine the quantity bounded. We therefore distinguish three observations: the production failure, a saved-matrix direction witness, and a coefficient-only obstruction for a precisely frozen majorant.

The coefficient obstruction fixes all variables other than the coefficient matrix. It is not an obstruction after changing the Young parameter, physical error budgets, observation/trial spaces, action approximants, trial-Gram upper bound or response inequality. The data/source commit in Appendix B fixes the numerical meaning of these choices. All new numerical statements in this note are copied from the completed diagnostics; paper preparation did not repeat their arithmetic.

2 The operator and inherited premises

For $x \geq 0$ set

$$\kappa(x) = e^{x/2} \sum_{n\geq 1} (4z_n(x)^2 - 6z_n(x)) e^{-z_n(x)}, \qquad z_n(x) = \pi n^2 e^{2x}, \tag{2.1}$$

and extend evenly. Put $c(x) = \cosh(x/2)$, $\mathrm{d}\nu = 2\kappa c\,\mathrm{d}x$, and $\mathcal{H} = L^2_{\mathrm{even}}(\mathbb{R}, \nu; \mathbb{C})$, with $\nu(\mathbb{R}) = 1$. Inner products are conjugate-linear in the first variable. The symmetric half measure is

$$\mathrm{d}\mu(s) = \frac{1}{2}\frac{e^{-|s|/2}}{1 - e^{-2|s|}}\,\mathrm{d}s + \frac{1}{2}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}(\delta_{\log n} + \delta_{-\log n}). \tag{2.2}$$

The singular continuous part is always used with the difference in

$$\mathcal{E}(f) = \iint \kappa(x)\kappa(z)|f(x) - f(z)|^2\,\mathrm{d}\mu(x - z)\,\mathrm{d}x. \tag{2.3}$$

Here the z integral includes the translated atoms. The two equal half contributions give the usual positive-shift formula, with no extra factor of two. Let $(\mathcal{E}, \mathcal{D})$ be the minimum closure of the compact smooth even core in the $\mathcal{E} + \mathcal{H}$ norm. With its representing operator A, define

$$L = A - \tfrac{1}{2}I + \tfrac{1}{2}P_1, \qquad \mathrm{D}(L) = \mathrm{D}(A), \qquad P_1 f = \langle 1, f\rangle_{\mathcal{H}}\,1. \tag{2.4}$$

No minimum/maximal-domain identification is used.

Hypothesis 2.1 (Inherited construction and physical inclusions)**.** The following are retained premises, not independently reconstructed by this note or by its finite checker.

1. The actual infinite theta source has the stated positivity, normalization, strip holomorphy, transform and global bounds. The minimum operator realization, compact-core signed identity, norm convergence of the sandwiched prime series, and justified weak source cancellation with the differentiated pole are those of the archived construction (A1–A4).

2. The whole-cell symbol, source and moment cover inclusions and the weighted omitted-prime bound give the auxiliary coercivity stated below (N1).

3. All 68 true trials belong to the minimum operator domain. Their whole-line action approximants, including affine exteriors, and all trial/action norm bounds are valid (N2).

4. The retained interval Grams enclose the actual physical inner products, including integration, truncation, source, transfer and serialization errors with their recorded ownership. This includes the recovered full 68-dimensional trial Gram (N3).

The source-generated space $\mathcal{N}$ is the closed span of 1 and $\kappa^{(2j)}/\kappa - 4^{-j}$, $j \geq 1$. The inherited inclusion $\mathcal{N} \subset \ker L$ is retained, never equality. The finite response argument here supplies no new energy estimate exploiting every $\mathcal{N}$-orthogonality condition. File identity is not a proof of any analytic or physical premise above.

3 The frozen response model

Write $O : \mathbb{C}^{372} \to \mathcal{H}$ for the archived observation synthesis (called W in the source). Its order consists of three Cholesky columns for each of 124 retained cells of width 1/2. If c_s, v_s are the saved cell centers and upper weights, put $a_s = \sqrt{v_s/\pi}$ and

$$\begin{aligned} Q_{3s}(x) &= a_s(1 - x^2/96)\cos(c_s x), \\ Q_{3s+1}(x) &= -a_s x \sin(c_s x)/(4\sqrt{3}), \\ Q_{3s+2}(x) &= -a_s x^2\cos(c_s x)/(48\sqrt{5}), \qquad w_j = Q_j/(2c). \end{aligned} \tag{3.1}$$

The fixed trial map $V : \mathbb{C}^{68} \to \mathrm{D}(L)$ consists of the original $V_{64} = OT_{64}$ and four admitted residual-source trials. Those four trials are not the translated null vectors often denoted q_t. The saved T_{64} and residual-source definitions and the entire observation order are unchanged.

The action anchor and the tested shift are

$$\delta_0 = 1/64, \quad \delta = h = 1/128, \quad H_0 = L + \delta_0 I + OO^*, \quad H = H_0 - hI. \tag{3.2}$$

The fixed full-line G approximates H_0V, not the action at the most recent successful shift. Define

$$G_h = G - hV, \qquad E = H_0V - G = HV - G_h. \tag{3.3}$$

The old 64 action columns retain their affine exterior; the four saved new action proxies have a charged zero-exterior approximation error. The true trials themselves are not replaced by compactly truncated proxies.

Use the fixed bounds

$$\|V\| \le \nu, \quad E^*E \preceq \sigma^2 I, \quad \|E\| \le \bar{\sigma}, \quad \sigma^2 = \sum_{i=1}^{68} B_i^2 \le \bar{\sigma}^2, \quad \tau = 1/100. \tag{3.4}$$

All 64 old and four corrected new B_i remain fixed. These are bounds, not measured actual norms. Each new-column bound already owns its Gamma, prime, exterior, source-transfer, frequency, coefficient, frame and pole terms; frame and pole are not additional error columns. The exact $\nu, \sigma^2, \bar{\sigma}$ are in the committed scalar and norm files.

Set

$$\begin{aligned} d_w &= \frac{1989293801}{35118080000000000}, \\ \epsilon_{\text{cover}} &= \frac{14194233101692862490101543299783481485}{3284055810247577849424937639623597936869376}, \\ F_w &= d_w + \epsilon_{\text{cover}}, \qquad a = 1/128 - F_w > 0. \end{aligned} \tag{3.5}$$

The retained cover/weighted-tail premise gives $H \succeq aI$. The original positive cover error has not been removed.

Let

$$M = O^*O, \; D = V^*O, \; K = G^*O, \; F_g = \operatorname{sym}(V^*G), \; N_g = G^*G, \; P = V^*V, \tag{3.6}$$

where $\operatorname{sym} X = (X + X^*)/2$. Thus N_g is not $\mathcal{N}$, and the file `E68.json` represents F_g, not the action error E. The full trial-Gram enclosure $|P - P_c| \le R_P$ gives the fixed exact matrix

$$P_{\text{up}} = P_c + \operatorname{diag}\Big(\sum_j (R_P)_{ij}\Big), \qquad P \preceq P_{\text{up}} \preceq \nu^2 I. \tag{3.7}$$

Its physical allowance is charged once. The radius of its subsequent 160-bit encoding represents only encoding error.

Define the exact physical quantities

$$\begin{aligned} &\alpha = (1+\tau)/a, && \beta = \nu\bar{\sigma} + (1 + 1/\tau)\sigma^2/a, \\ &S = \alpha N_g - (1 + 2\alpha h)F_g + (h + \alpha h^2)P_{\text{up}} + \beta I, && B = \alpha K - (1 + \alpha h)D, \end{aligned} \tag{3.8}$$

and for every 68×372 coefficient matrix C' put

$$U(C') = \alpha M - B^*C' - C'^*B + C'^*SC'. \tag{3.9}$$

The frozen construction in the theorem means exactly (3.2)–(3.9), its physical input enclosures and the same observation/trial/action definitions.

Lemma 3.1 (Response majorant and the normal lower bound)**.** *Under the inherited premises,*

$$O^*H^{-1}O \preceq U(C') \quad \textit{for every } C', \qquad S \succeq m_h I, \tag{3.10}$$

where

$$m_h = \beta - \frac{\sigma^2}{4\tau a} = \nu\bar{\sigma} + 76\sigma^2/a > 0. \tag{3.11}$$

Proof. For $Y = VC'$ and $T = O - HY$, the operator-domain response identity is

$$O^*H^{-1}O = O^*Y + Y^*O - Y^*HY + T^*H^{-1}T.$$

Write $R = O - G_hC'$, so $T = R - EC'$. Bound H^{-1} by $a^{-1}I$, apply Young's inequality with parameter τ, use (3.4), and bound $-\operatorname{sym}(V^*E)$ by $\nu\bar{\sigma}I$. Before replacing P, this yields

$$C'^*D + D^*C' - C'^*F_gC' + hC'^*PC' + \alpha R^*R + \beta C'^*C'.$$

Expanding R^*R shows that replacing the collected P term by P_{up} adds $(h+\alpha h^2)C'^*(P_{\mathrm{up}}-P)C' \succeq 0$, giving (3.9).

For the lower bound, let $u = Vz$, $g = G_hz$. Since $Hu = g + Ez$, $\|u\| \le (\|g\| + \sigma\|z\|)/a$. Consequently

$$z^*[\alpha G_h^*G_h - \operatorname{sym}(V^*G_h) + \beta I]z \ge \frac{\tau}{a}\left(\|g\| - \frac{\sigma\|z\|}{2\tau}\right)^2 + m_h\|z\|^2.$$

The matrix S is the bracketed matrix plus $(h + \alpha h^2)(P_{\mathrm{up}} - P) \succeq 0$. □

4 The failure direction and three different matrices

The one production attempt made one new coefficient proposal C, rounded once to exact80-bit dyadics. Its full372 factor stopped at zero-based pivot210. This did not yet prove that the saved matrix was indefinite, nor exclude other C'.

A subsequent diagnostic constructed one80-bit dyadic direction on coordinates $0, \dots, 210$, with last coordinate1 and161 exact zero coordinates appended. Its exact quadratic checker verified

$$v^*(I - U_{\mathrm{mid}})v < 0, \qquad v^*(I - U_{\mathrm{cert}})v < 0. \tag{4.1}$$

Here U_{mid} is the saved100-bit `U_MODEL` center, and $U_{\mathrm{cert}} = U_{\mathrm{mid}} + P_{\mathrm{row}}$ uses all372 columns in every row-radius sum. Thus removing the final pad alone does not remove this saved-center failure. This statement concerns stored upper-bound matrices, not the sign of the true L.

For the coefficient argument distinguish:

1. the exact physical $U(C')$ of (3.9);
2. the nominal polynomial $Q_0(C')$ obtained from the fixed exact 160-bit dyadic centers α_0, M_0, B_0, S_0;
3. the separately rounded stored center U_{mid} and its padded certificate at the existing C.

In particular α_0 is a scalar-ball center and need not equal exact α. Entrywise requantization means $U_{\mathrm{mid}} \neq Q_0(C)$ in general. The exact check retained

$$d_{\mathrm{round}} = v^*U_{\mathrm{mid}}v - v^*Q_0(C)v, \qquad |d_{\mathrm{round}}| \le 2^{-100}\|v\|_1^2. \tag{4.2}$$

For completeness, the320-to100-bit endpoint quantizer maps a raw center to either its lower or upper100-bit neighboring point: writing the raw center and radius modulo 2^{220} verifies this

directly. Hence each center error is at most 2^{-100}, which gives (4.2). This only controls final center encoding; it removes no physical radius.

Let $c = Cv$, $n = v^*v$ and

$$r_0 = B_0v - S_0c, \qquad e_0 = v^*Q_0(C)v - n. \tag{4.3}$$

The saved normal residual stores $S_0C - B_0$, with the opposite sign. The check independently compared its contraction with $-r_0$ at the correct scale. Solve inaccuracy and coefficient rounding are already represented in r_0; there is no second coefficient-error allowance.

5 A bound on every coefficient-only improvement

Lemma 5.1 (Elementary residual bound)**.** *Let $S \succeq mI > 0$, $v \neq 0$, $b = Bv$ and $c = Cv$. For*

$$q(y) = \alpha v^*Mv - 2\Re(b^*y) + y^*Sy, \qquad r = b - Sc,$$

one has

$$q(c) - \inf_y q(y) = r^*S^{-1}r \leq \|r\|^2/m. \tag{5.1}$$

If $q(c) - n > \|r\|^2/m$, every coefficient matrix C' fails $v^*U(C')v < n$.

Proof. Completing the square gives

$$q(y) = \alpha v^*Mv - b^*S^{-1}b + (y - S^{-1}b)^*S(y - S^{-1}b).$$

At $y = c$ the square is $r^*S^{-1}r$, bounded by $\|r\|^2/m$. Every y is attained by $C' = C + (y-c)v^*/n$, so the coefficient quantifier is unrestricted. □

This is a bound on the greatest possible decrease, not merely an error estimate for one linear solver. The lemma is standard finite-dimensional square completion. Its contribution here is its scoped application with the correct enclosures and exact comparison evidence.

If the physical S lies in $S_0 \pm R_S$, put $\rho_S = \max_i \sum_j (R_S)_{ij}$. Then

$$S_0 \succeq (m_h - \rho_S)I. \tag{5.2}$$

The executed check validated the scalar expression, all68 radius rows and the raw-input enclosure bridge. The direct center Gershgorin lower bound g_0 was negative. It therefore used a downward160-bit dyadic $m_0 \leq m_h - \rho_S$, with the inherited premises explicitly retained. This is not a premise-free positive-definiteness proof for S_0. An arbitrary matrix in the full interval box has a different obligation, for example $m_h - 2\rho_S$; no whole-box exclusion is claimed.

For the physical comparison, let R_B, R_M be the primitive enclosing radii and $|\alpha - \alpha_0| \leq r_\alpha$. At the same c, v, define

$$\begin{aligned} t &= R_B|v| + R_S|c|, \\ \epsilon_q &= |v|^T(|\alpha_0|R_M + r_\alpha|M_0| + r_\alpha R_M)|v| + 2|c|^T R_B|v| + |c|^T R_S|c|, \\ R_+ &= \sum_i (|(r_0)_i| + t_i)^2. \end{aligned} \tag{5.3}$$

Then $|q_{\text{actual}}(c) - q_0(c)| \leq \epsilon_q$ and $\|r_{\text{actual}}\|^2 \leq R_+$. With a valid physical lower bound $S \succeq m_{\text{actual}}I > 0$, Lemma 5.1 gives

$$\inf_{C'} v^*U(C')v - n \geq e_0 - \epsilon_q - R_+/m_{\text{actual}}. \tag{5.4}$$

No `U_MODEL` radius or final row pad is added again to (5.3). The old coefficient-dependent pad is not fixed across C'. The physical admissible set is not identified with the unrestricted interval box.

6 The two conditional exclusions

The accepted exact report records

$$e_0 > \|r_0\|^2/m_0, \qquad e_0 - \epsilon_q > R_+/m_{\mathrm{actual}}. \tag{6.1}$$

All products use the same v. No observed excess is normalized separately from the improvement bound. The exact lower-bound encodings are

$$\begin{aligned} m_0 &= \frac{11908439363740784705638882337283564911265}{2^{160}}, \\ m_{\mathrm{actual}} &= \frac{12019713510740312408761598061526876985806}{2^{160}}. \end{aligned} \tag{6.2}$$

The first comes from (5.2); the second is a downward encoding of (3.11). These values are existing report fields, not newly evaluated approximations.

Theorem 6.1 (Conditional obstruction in the frozen family)**.** *Assume the inherited analytic and physical inclusions and use the commit-pinned data of this note. Keep the same v, $\delta = h = 1/128$, $\tau = 1/100$, $O_{372}, V_{68}, G, P_{\mathrm{up}}$, norm/action-error bounds, primitive enclosures and majorant construction. Then:*

1. *Every unrestricted coefficient matrix C' satisfies $v^*Q_0(C')v > v^*v$ for the fixed exact nominal polynomial.*
2. *Every unrestricted coefficient matrix C' satisfies $v^*U(C')v > v^*v$ for the specified physical-majorant family under its retained premises.*

The nominal conclusion also retains those premises through its lower bound m_0. Neither conclusion determines the sign of L.

Proof. Apply Lemma 5.1 to the nominal polynomial using the first strict comparison in (6.1). Apply (5.4) and the second strict comparison for the physical statement. The exact integer form of both comparisons is recorded below and in the public report. The coefficient quantifier follows from the final step of Lemma 5.1. □

Write the saved rational values as

$$e_0 = E_0/2^{480}, \quad \epsilon_q = A_q/2^{480}, \quad \|r_0\|^2 = Z/2^{640}, \quad R_+ = P_+/2^{640},$$

and $m_0 = m_0^{\#}/2^{160}$, $m_{\mathrm{actual}} = m_a^{\#}/2^{160}$. The two exact integer margins are

$$J_0 = E_0 m_0^{\#} - Z > 0, \qquad J_a = (E_0 - A_q)m_a^{\#} - P_+ > 0. \tag{6.3}$$

They give the explicit lower bounds $J_0/(m_0^{\#}2^{480})$ and $J_a/(m_a^{\#}2^{480})$ for the respective optimized excesses. Appendix A prints the accepted margin integers without decimal approximation; the committed JSON supplies every operand and scale.

The separate sufficient test for existence of a repairing direction, using an exact row upper bound L_0 for S_0, was inconclusive. That failure of a sufficient test does not itself prove an obstruction; the two independent strict comparisons above do.

7 Execution evidence and reproducibility

The three runs must not be merged into one success narrative.

Run	Host exit	Resource accepted	Full wall (s)
Midpoint128 production	2	false	not certified
Same-direction diagnostic	0	true	1.156969134
Coefficient-gap diagnostic	0	true	2.343085096

The failed production made one coefficient solve, one full372 consumer and one factor attempt, stopping at pivot210. It reached no independent scientific checker and produced no accepted complete timing/RSS witness. Its resource-acceptance flag remains false. Reaping observed on that failure path is not retroactively a successful external resource certificate.

The first subsequent diagnostic used one210-dimensional direction solve with one right-hand side and one exact same-direction check. Its observed worker peak RSS was77,244KiB. The coefficient-gap diagnostic used one standard-library exact checker and no solve, factor, inverse, scientific native import, full372 consumer, old diagnostic replay or source/action/ FFT/integral call. Its observed peak RSS was65,100KiB. Both successful diagnostics have actual exit0 and external completed-reap/no-descendant evidence. Their one-CPU elapsed-capacity upper bounds are not sums of measured process CPU usage. These are existing execution records, not measurements repeated during publication.

The coefficient check used full68-row normal/enclosure checks and selected68-by211 and211-by211 contractions. The remaining161 entries of v are exactly zero. Input scales160, coefficient/witness scales80 and natural exact product scales up to640 describe denominators, not a floating-precision increase. The implementation enforces dense integer guards and a separate bounded scalar-rational parser.

The public distribution preserves27 scientific inputs, the byte-identical executed coefficient checker, relevant producer-formula source, the exact reports, and a curated execution summary. Private conversation, account, signed-URL and authorization/environment records are excluded. The unchanged worker report retains its provisional resource flag; the later externally accepted outcome is identified separately.

A portable reader adapter supplies relative paths and directly rechecks the prior same-vector quadratic values before using the unchanged coefficient arithmetic. That adapter was statically inspected but *not numerically executed during publication.* It is not a new historical run, a hermetic physical-production environment, or a replay of the original OS/resource protocol. The reproduction instructions distinguish byte checks, optional reader-side saved arithmetic, and the inherited physical input-generation obligations.

8 Limits and unevaluated follow-up work

An upper-majorant obstruction does not give a negative direction for $O^*H^{-1}O$ or for L. In particular, it does not exclude another method at $\delta = 1/128$, a changed τ, a different trial/observation map, sharper source/error information, or a different majorant. No assertion about all pairs (C', τ) is made.

A separate reading audit identifies norm, Young, inverse-replacement, trial-Gram and enclosure allowances. It has not measured their realized directional sizes or identified a dominant loss. A possible next test is to keep the current C, v and vary only τ, using scalar contractions of the saved primitive data. No such numerical test has been run, no successful parameter has been found, and neither that proposal nor the loss audit is a theorem or numerical conclusion of this paper. Directional inverse-response bounds and genuine trial expansion remain other untested possibilities requiring additional information.

The argument does not reconstruct the source premises, establish $\mathcal{N} = \ker L$, prove full-$\mathcal{N}^\perp$ coercivity, or identify a maximal operator domain. Internal independent audits are not academic peer review or formal proof-assistant verification. No literature-priority claim is made. The status remains `RH_PROOF=NOT-OBTAINED`.

A Exact recorded positive margins

The following digit blocks are consecutive pieces of each positive integer in (6.3); concatenate the lines without spaces. They are copied as text from the accepted exact-comparison record.

J_0.

```
366420405067584375457281240633757545915869894524991627717070606 1
612233132836010025733104747614923100844076093926401030122921526 4
210510536506001383221540312364810041609543175762786915817
```

J_a.

```
369844287720436065297831866662575243640460844307205979467456501 7
122379747225449699892202005126375589330092919866609348006172220 6
117896686399102888331747672406861701864645557827732858950
```

The corresponding complete data record has SHA256

`7274b6e56be6c00f03a94d2acb6fcb9da30c30606edc203c3cf05c80b63cfa33`.

B Commit-pinned public evidence

The data and source commit [3] is

`f61c7af23c7e4027cc3cb768faa40273f1146ffd`.

The artifact directory is

`https://github.com/serabing-hash/theta-weighted-certificate/tree/f61c7af23c7e4027cc3cb768faa40273f1146ffd/research/2026-10-18-coefficient-obstruction`

This pins the scientific evidence before the later manuscript commit. Direct fixed links identify the exact comparisons, unchanged coefficient report, directional center/pad report, and separated execution summary. The full saved-data distribution is also available as a commit-pinned artifact ZIP.

The exact executed checker SHA256 is

`fe758654a469ae37f446b99ee329cca67034ac9499ec01fa02cb08c840d51a32`.

The saved witness SHA256 is

`9f659a0426dcf7f1e25f9602daae30bdf5d7d16c766fc082c296bf676d996fcb`.

The accepted coefficient-check plan SHA256 is

`0369628cf7ce778a8c32d6546eb550b27eddd2690d59a763ea23bc9635606ee0`.

The public `INPUTS.json`, `MANIFEST.sha256`, `ASSUMPTIONS.md` and `REPRODUCTION.md` specify the remaining identities and limitations. Hashes establish byte identity, not the physical truth of an inclusion.

References

[1] J. Choi, *Fixed-threshold source audit and corrected full372 public evidence*, research archive,8 October2026. https://github.com/serabing-hash/theta-weighted-certificate/releases/tag/v2026.10.08-fixed-threshold-source-audit. Used with its explicit analytic and physical-inclusion premises.

[2] J. Choi, *A Conditional Improvement of a Fixed Theta Operator Lower Bound*, research note,18 October2026. https://github.com/serabing-hash/theta-weighted-certificate/blob/888126c019130be9693e004bcd9984e5a1540f32/papers/2026-10-18-fixed-threshold-shift/fixed_threshold_shift.tex.

[3] J. Choi, *Conditional coefficient-obstruction source and saved-data evidence*, research artifacts,18 October2026, commit `f61c7af23c7e4027cc3cb768faa40273f1146ffd`. See Appendix B for fixed links and reproduction boundaries.

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